Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Contents

Integrals

1.1 Easy Integral

Trigonometric Definite Integral

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos(x)}{2-\cos^2(x)}\,dx$$

Solution: We solve this using a simple u-substitution. First, we rewrite the denominator using the Pythagorean identity $\cos^2(x) = 1 - \sin^2(x)$:

$$2-\cos^2(x) = 2-(1-\sin^2(x)) = 1+\sin^2(x)$$

The integral becomes:

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos(x)}{1+\sin^2(x)}\,dx$$

Let $u = \sin(x)$. Then, the differential is $du = \cos(x)\,dx$. Next, we update our limits of integration: When $x = -\frac{\pi}{2}$, $u = \sin(-\frac{\pi}{2}) = -1$. When $x = \frac{\pi}{2}$, $u = \sin(\frac{\pi}{2}) = 1$. Substituting these into the integral yields:

$$\int_{-1}^{1} \frac{1}{1+u^2}\,du$$

This is a standard integral evaluating to the arctangent function:

$$[\arctan(u)]_{-1}^{1} = \arctan(1) - \arctan(-1)$$

Since $\arctan(1) = \frac{\pi}{4}$ and $\arctan(-1) = -\frac{\pi}{4}$:

$$\frac{\pi}{4} - \left(-\frac{\pi}{4}\right) = \frac{\pi}{2}$$

1.2 Medium Integral

Maclaurin Series Approximation

Approximate

$$\int_0^1 x^2 \sin(x^2)\,dx$$

to 2 decimal places.

Solution: Because the antiderivative of this function is not expressible in terms of elementary functions, we utilize the Maclaurin series expansion for $\sin(u)$:

$$\sin(u) = u - \frac{u^3}{3!} + \frac{u^5}{5!} - \dots$$

Substitute $u = x^2$:

$$\sin(x^2) = x^2 - \frac{x^6}{6} + \frac{x^{10}}{120} - \dots$$

Multiply the entire series by x^2:

$$x^2\sin(x^2) = x^4 - \frac{x^8}{6} + \frac{x^{12}}{120} - \dots$$

Now, integrate this series term-by-term from 0 to 1:

$$\int_0^1 \left(x^4 - \frac{x^8}{6} + \frac{x^{12}}{120} - \dots \right) dx = \left[\frac{x^5}{5} - \frac{x^9}{54} + \frac{x^{13}}{1560} - \dots \right]_0^1$$
$$= \frac{1}{5} - \frac{1}{54} + \frac{1}{1560} - \dots$$

Convert these fractions to decimals to find the approximation:

$$\frac{1}{5} = 0.2000$$
$$\frac{1}{54} \approx 0.0185$$
$$\frac{1}{1560} \approx 0.0006$$

Summing the first three terms:

$$0.2000 - 0.0185 + 0.0006 = 0.1821$$

Rounding to 2 decimal places, the approximation is **0.18**.

1.3 Hard Integral

Evaluating an Integral with Fourier Series

Evaluate the Integral:

$$I = \int_0^{2\pi} \frac{\ln(2\sin(x/2))}{\frac{5}{4} - \cos(x)}\, dx$$

Solution: We can expand the numerator using the known Fourier series expansion for the logarithmic function. For $x \in (0, 2\pi)$:

$$\ln(2\sin(x/2)) = -\sum_{n=1}^{\infty} \frac{\cos(nx)}{n}$$

Substitute this series into the integral:

$$I = \int_0^{2\pi} \frac{-\sum_{n=1}^{\infty} \frac{\cos(nx)}{n}}{\frac{5}{4} - \cos(x)}\, dx = -\sum_{n=1}^{\infty} \frac{1}{n} \int_0^{2\pi} \frac{\cos(nx)}{\frac{5}{4} - \cos(x)}\, dx$$

We use the standard known integral formula for $\int_0^{2\pi} \frac{\cos(nx)}{a - \cos(x)}\, dx$, where $a > 1$:

$$\int_0^{2\pi} \frac{\cos(nx)}{a - \cos(x)}\, dx = \frac{2\pi}{\sqrt{a^2 - 1}} \alpha^n$$

where $\alpha = a - \sqrt{a^2 - 1}$. In our case, $a = \frac{5}{4}$.

$$\sqrt{a^2 - 1} = \sqrt{\frac{25}{16} - 1} = \sqrt{\frac{9}{16}} = \frac{3}{4}$$

$$\alpha = \frac{5}{4} - \frac{3}{4} = \frac{2}{4} = \frac{1}{2}$$

The integral of the cosine term thus evaluates to:

$$\frac{2\pi}{3/4}\left(\frac{1}{2}\right)^n = \frac{8\pi}{3}\left(\frac{1}{2}\right)^n$$

Substitute this back into the sum:

$$I = -\sum_{n=1}^{\infty}\frac{1}{n}\left(\frac{8\pi}{3}\left(\frac{1}{2}\right)^n\right) = -\frac{8\pi}{3}\sum_{n=1}^{\infty}\frac{(1/2)^n}{n}$$

Recall the Maclaurin series for the natural logarithm: $\sum_{n=1}^{\infty}\frac{x^n}{n} = -\ln(1-x)$. Let $x = \frac{1}{2}$:

$$\sum_{n=1}^{\infty}\frac{(1/2)^n}{n} = -\ln\left(1-\frac{1}{2}\right) = -\ln\left(\frac{1}{2}\right) = \ln(2)$$

Therefore, the final value of the integral is:

$$I = -\frac{8\pi\ln(2)}{3}$$

Differentials

2.1 Medium Differential

Linear Second-Order ODE with Golden Ratio

Let $\varphi = \frac{1+\sqrt{5}}{2}$ be the golden ratio. A twice-differentiable function $y(x)$ satisfies the following differential equation:

$$y''(x) + y'(x) - y(x) = x$$

If $\int_0^\infty (y(x) + x + 1)\, dx = \varphi - 1$, determine the value of $y(1 - \varphi)$.

Solution: First, find the homogeneous solution to $y'' + y' - y = 0$. The characteristic equation is $r^2 + r - 1 = 0$. By the quadratic formula:

$$r = \frac{-1 \pm \sqrt{1 - 4(1)(-1)}}{2} = \frac{-1 \pm \sqrt{5}}{2}$$

Notice that the roots correspond beautifully to the golden ratio: $r_1 = \varphi - 1$ (which is > 0) and $r_2 = -\varphi$ (which is < 0).

$$y_h(x) = C_1 e^{(\varphi-1)x} + C_2 e^{-\varphi x}$$

Next, we find a particular solution $y_p(x) = Ax + B$. Substituting this into the ODE:

$$0 + A - (Ax + B) = x \implies -Ax + (A - B) = x$$

Matching coefficients gives $A = -1$ and $A - B = 0 \implies B = -1$.

$$y(x) = C_1 e^{(\varphi-1)x} + C_2 e^{-\varphi x} - x - 1$$

We are given the integral condition:

$$\int_0^\infty (y(x) + x + 1)\, dx = \int_0^\infty \left(C_1 e^{(\varphi-1)x} + C_2 e^{-\varphi x}\right) dx = \varphi - 1$$

For this improper integral to converge, the coefficient of the exponential with a positive growth rate must be zero. Since $\varphi - 1 > 0$, we must have $C_1 = 0$. Evaluating the converging part:

$$\int_0^\infty C_2 e^{-\varphi x}\, dx = \left[-\frac{C_2}{\varphi} e^{-\varphi x}\right]_0^\infty = \frac{C_2}{\varphi}$$

Set this equal to the given value:

$$\frac{C_2}{\varphi} = \varphi - 1 \implies C_2 = \varphi(\varphi - 1) = \varphi^2 - \varphi$$

Because φ is the golden ratio, it satisfies $\varphi^2 - \varphi - 1 = 0$, meaning $C_2 = 1$. The specific function is:

$$y(x) = e^{-\varphi x} - x - 1$$

Now, evaluate at $x = 1 - \varphi$:

$$y(1 - \varphi) = e^{-\varphi(1-\varphi)} - (1 - \varphi) - 1$$

Simplify the exponent: $-\varphi(1 - \varphi) = -\varphi + \varphi^2 = 1$.

$$y(1 - \varphi) = e^1 - 1 + \varphi - 1 = e + \varphi - 2$$

2.2 Hard Differential

Cauchy-Euler Equation

The temperature anomaly $T(x)$ inside a fusion reactor wall is measured at depth x metres. The wall is designed so that the heat flux obeys the Cauchy-Euler relationship:

$$x^2T''(x) - 5xT'(x) + 9T(x) = x^3 \ln(x)$$

Given the boundary conditions $T(1) = 4$ and $T'(1) = 7$, find the anomaly at depth $x = e$.

Solution: This is a non-homogeneous Cauchy-Euler equation. We solve the homogeneous part $x^2T'' - 5xT' + 9T = 0$ by assuming $T(x) = x^r$:

$$r(r-1) - 5r + 9 = 0 \implies r^2 - 6r + 9 = 0 \implies (r-3)^2 = 0$$

Since $r = 3$ is a repeated root, the homogeneous solution is $T_h(x) = (C_1 + C_2 \ln x)x^3$. To find the particular solution, we use the substitution $x = e^t \implies t = \ln x$. The equation transforms to:

$$\frac{d^2T}{dt^2} - 6\frac{dT}{dt} + 9T = te^{3t}$$

We use the method of undetermined coefficients. The RHS is te^{3t}, and 3 is a double root of the characteristic equation. Our trial solution is $T_p(t) = t^2(At + B)e^{3t} = (At^3 + Bt^2)e^{3t}$. Differentiating and substituting this into the ODE, we match coefficients to get $A = \frac{1}{6}$ and $B = 0$.

$$T_p(t) = \frac{1}{6}t^3e^{3t}$$

Reverting to x, $T_p(x) = \frac{1}{6}x^3(\ln x)^3$. The general solution is:

$$T(x) = C_1x^3 + C_2x^3 \ln x + \frac{1}{6}x^3(\ln x)^3$$

Apply the boundary conditions: $T(1) = C_1(1) + 0 + 0 = 4 \implies C_1 = 4$. Take the derivative $T'(x)$:

$$T'(x) = 12x^2 + C_2(3x^2 \ln x + x^2) + \frac{1}{6}\left(3x^2(\ln x)^3 + 3x^2(\ln x)^2\right)$$

$T'(1) = 12 + C_2 = 7 \implies C_2 = -5$. The specific solution is:

$$T(x) = 4x^3 - 5x^3 \ln x + \frac{1}{6}x^3(\ln x)^3 = x^3\left(4 - 5\ln x + \frac{1}{6}(\ln x)^3\right)$$

Evaluate at $x = e$:

$$T(e) = e^3\left(4 - 5(1) + \frac{1}{6}(1)^3\right) = e^3\left(-1 + \frac{1}{6}\right) = -\frac{5}{6}e^3$$

Derivatives

3.1 Medium Derivative

Complex Chain Rule Evaluation

Let $f(x) = \sqrt[6]{\cos(\ln(3x+8))} \cdot \sinh^2(\ln(x+2))$. Find $f'(-1.3)$.

Solution: We apply the product rule and chain rule to differentiate the function $f(x) = u(x)v(x)$, where: $u(x) = [\cos(\ln(3x+8))]^{1/6}$ $v(x) = \sinh^2(\ln(x+2))$

Differentiating $u(x)$:

$$u'(x) = \frac{1}{6}[\cos(\ln(3x+8))]^{-5/6} \cdot (-\sin(\ln(3x+8))) \cdot \frac{3}{3x+8}$$

Differentiating $v(x)$:

$$v'(x) = 2\sinh(\ln(x+2))\cosh(\ln(x+2)) \cdot \frac{1}{x+2}$$

By the product rule $f'(x) = u'(x)v(x) + u(x)v'(x)$:

$$f'(x) = \left[-\frac{1}{2(3x+8)} \frac{\sin(\ln(3x+8))}{[\cos(\ln(3x+8))]^{5/6}}\right] \sinh^2(\ln(x+2)) + [\cos(\ln(3x+8))]^{1/6} \left[\frac{2\sinh(\ln(x+2))\cosh(\ln(x+2))}{x+2}\right]$$

We now evaluate this expression specifically at $x = -1.3$: First, find the inner terms at $x = -1.3$: $3x+8 = 3(-1.3)+8 = -3.9+8 = 4.1$ $x+2 = -1.3+2 = 0.7$

Substitute these back into the rigorous analytic derivative format:

$$f'(-1.3) = \left[-\frac{1}{8.2} \frac{\sin(\ln(4.1))}{[\cos(\ln(4.1))]^{5/6}}\right] \sinh^2(\ln(0.7)) + [\cos(\ln(4.1))]^{1/6} \left[\frac{2\sinh(\ln(0.7))\cosh(\ln(0.7))}{0.7}\right]$$

3.2 Hard Derivative

Reconstructing a Function from a Gradient

A function $f(x,y)$ with parameters a, b has the gradient $\nabla f = (3x^2 + by^2)\hat{\imath} + (2bxy + a)\hat{\jmath}$ and satisfies $\frac{\partial^3 f}{\partial x^3} + \frac{1}{x}\frac{\partial^2 f}{\partial y^2} = 10$. Knowing that $f(0,0) = 0$ and $f(1,1) = 2$, find $f(1,2)$.

Solution: We use the gradient to find the general form of $f(x,y)$. From $f_x = 3x^2 + by^2$, we integrate with respect to x:

$$f(x,y) = x^3 + bxy^2 + g(y)$$

Differentiate this with respect to y and equate to the given f_y:

$$f_y = 2bxy + g'(y) = 2bxy + a \implies g'(y) = a \implies g(y) = ay + C$$

So, $f(x,y) = x^3 + bxy^2 + ay + C$. Using the condition $f(0,0) = 0 \implies C = 0$.

Next, we evaluate the mixed partials equation: $\frac{\partial^3 f}{\partial x^3} + \frac{1}{x}\frac{\partial^2 f}{\partial y^2} = 10$. $f_{xx} = 6x \implies f_{xxx} = 6$. $f_y = 2bxy + a \implies f_{yy} = 2bx$. Substitute these into the equation:

$$6 + \frac{1}{x}(2bx) = 10 \implies 6 + 2b = 10 \implies 2b = 4 \implies b = 2$$

Our function is now $f(x, y) = x^3 + 2xy^2 + ay$. We use the second coordinate condition $f(1, 1) = 2$ to find a:

$$f(1, 1) = 1^3 + 2(1)(1)^2 + a(1) = 2 \implies 1 + 2 + a = 2 \implies 3 + a = 2 \implies a = -1$$

The exact function is $f(x, y) = x^3 + 2xy^2 - y$. Finally, we evaluate $f(1, 2)$:

$$f(1, 2) = 1^3 + 2(1)(2)^2 - 2 = 1 + 8 - 2 = 7$$

Physics & Engineering

4.1 Circuits

RC Circuit Transient

A capacitor of capacitance $C = 2\,\mu\text{F}$ is connected in series with a resistor $R = 500\,\text{k}\Omega$ and an ideal battery of emf $\mathcal{E} = 12$ V. The switch is closed at time $t = 0$. Find the charge on the capacitor in microcoulombs (μC) at time $t = 1$ s. Take $e \approx 2.7183$.

Solution: First, calculate the time constant τ of the RC circuit:

$$\tau = RC = (500 \times 10^3\,\Omega)(2 \times 10^{-6}\,\text{F}) = 1000 \times 10^{-3}\,\text{s} = 1\,\text{s}$$

The charge $Q(t)$ on a charging capacitor is given by:

$$Q(t) = C\mathcal{E}(1 - e^{-t/\tau})$$

The maximum charge Q_{max} is:

$$Q_{max} = C\mathcal{E} = (2\,\mu\text{F})(12\text{ V}) = 24\,\mu\text{C}$$

We are looking for the charge at $t = 1$ s, which perfectly matches one time constant τ:

$$Q(1) = 24(1 - e^{-1})$$

Using the approximation $e \approx 2.7183$:

$$e^{-1} = \frac{1}{2.7183} \approx 0.367877$$

$$Q(1) = 24(1 - 0.367877) = 24(0.632123) \approx 15.17\,\mu\text{C}$$

4.2 Mechanics

Elastic Collision and Spring Energy

A block A of mass $m_A = 1$ kg is connected to block B of mass $m_B = 2$ kg by a massless spring with constant $k = 200\text{ N m}^{-1}$. The system is at rest on a smooth surface. Block C of mass $m_C = 1$ kg moves at $v_C = 6\text{ m s}^{-1}$ and collides perfectly elastically with block A. Determine the maximum compression of the spring in meters.

Solution: Since block C and block A have the exact same mass (1 kg) and undergo a perfectly elastic collision, they instantly swap velocities. Block C comes to rest, and block A acquires the initial velocity of block C: $v_{A,initial} = 6\text{ m/s}$. We now treat A and B as an isolated system. Maximum compression occurs when blocks A and B reach the same common velocity v_f. By conservation of linear momentum:

$$m_A v_{A,initial} = (m_A + m_B)v_f \implies 1(6) = (1+2)v_f \implies 3v_f = 6 \implies v_f = 2\text{ m/s}$$

Next, apply conservation of mechanical energy to find the maximum compression x_{max}:

$$\frac{1}{2}m_A v_{A,initial}^2 = \frac{1}{2}(m_A + m_B)v_f^2 + \frac{1}{2}k x_{max}^2$$

Substitute the known values:

$$\frac{1}{2}(1)(6^2) = \frac{1}{2}(3)(2^2) + \frac{1}{2}(200)x_{max}^2$$

$$18 = 6 + 100x_{max}^2 \implies 12 = 100x_{max}^2$$

$$x_{max}^2 = \frac{12}{100} = 0.12$$

$$x_{max} = \sqrt{0.12} = \frac{2\sqrt{3}}{10} = \frac{\sqrt{3}}{5} \approx 0.346 \text{ m}$$

4.3 Quantum

Matter Waves

A proton and an alpha particle (He^{2+}) are both accelerated from rest through the exact same electrostatic potential difference V. Determine the ratio of the de Broglie wavelength of the proton to that of the alpha particle (λ_p/λ_α).

Solution: The kinetic energy K gained by a particle of charge q accelerated through a potential difference V is $K = qV$. The momentum p of the particle is related to its kinetic energy by $p = \sqrt{2mK} = \sqrt{2mqV}$. The de Broglie wavelength is given by $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mqV}}$. For the proton:

$$\lambda_p = \frac{h}{\sqrt{2m_p q_p V}}$$

For the alpha particle:

$$\lambda_\alpha = \frac{h}{\sqrt{2m_\alpha q_\alpha V}}$$

We want the ratio $\frac{\lambda_p}{\lambda_\alpha}$:

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{\frac{h}{\sqrt{2m_p q_p V}}}{\frac{h}{\sqrt{2m_\alpha q_\alpha V}}} = \sqrt{\frac{m_\alpha q_\alpha}{m_p q_p}}$$

An alpha particle consists of 2 protons and 2 neutrons. Therefore, its mass is approximately 4 times that of a proton ($m_\alpha \approx 4m_p$) and its charge is strictly twice that of a proton ($q_\alpha = 2q_p$). Substituting these relationships:

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{(4m_p)(2q_p)}{m_p q_p}} = \sqrt{8} = 2\sqrt{2}$$